# palmrungui

palmgroup

Institute of Meteorology and Climatology, Leibniz Universität Hannover

# Overview

- By default, PALM runs are started via linux shell command line:

```
palmrun -r example_cbl -a "d3#" […]
```

- Graphical user interface for palmrun can be used for simplified usage:

```
palmrungui
```

- palmrungui is automatically installed with palm (it requires additional packages: pyqt5, python3, see installation):
  - `pip install pyqt5`
  - `sudo apt-get install python3-pyqt5`
  - `...`
- Start palmrungui from any place via the shell (it might be required to install xterm):

```
palmrungui &
```

# palmrungui

palmrungui

Start Tools Help

Default | Advanced

**History**

Recently submitted jobs:

2023/08/08 15:21: run_B2_sub035v2 (default)
2023/08/08 15:21: run_B2_sub035_upgrade100v2 (default)

**Browser**

Browse at: /home/maronga/palm_23.04/JOBS/ Update (5 min)

Available jobs:
- run_B2_sub035_upgrade100v2
- run_B2_sub035v2

Input files:
- .directory (58 B)
- run_B2_sub035v2_p3d (6 KB)
- run_B2_sub035v2_p3dr (4 KB)

Output files:

User code:
- .directory
- land_surface_model_mod.f90
- radiation_model_mod.f90
- surface_mod.f90
- urban_surface_mod.f90

Monitoring files:

**Execution**

**Selected job:** run_B2_sub035v2

Configuration: default

Edit | (re-)Build | Full rebuild

Use pre-built source code

Cores: 4 Tasks per node: 4

Wallclock time (s): Queue:

Account: Wait for job:

Mode: Initial run

Allow restarts
Keep temporary data
Use cyclic fill method
Read sky view factors from file
Write sky view factors to file

palmrun | Silent mode | Clear | Quit

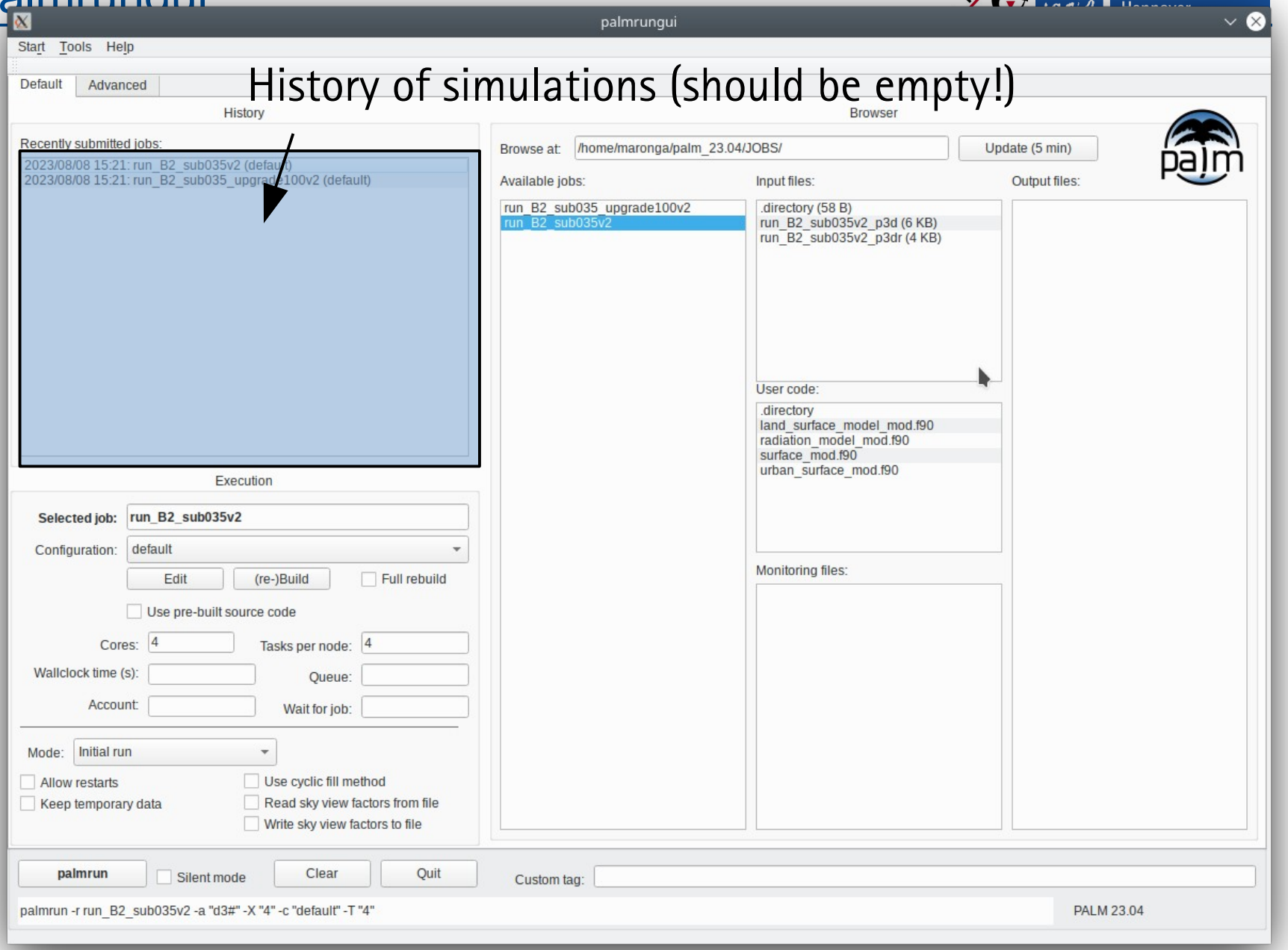
Custom tag:

palmrun -r run_B2_sub035v2 -a "d3#" -X "4" -c "default" -T "4"

PALM 23.04

# palmrungui

## History of simulations (should be empty!)



# Available PALM setups

palmrungui

Start Tools Help

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radiation_model_mod.f90
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urban_surface_mod.f90

Monitoring files:

**Execution**

**Selected job:** run_B2_sub035v2

Configuration: default

Edit | (re-)Build | Full rebuild

Use pre-built source code

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Wallclock time (s): Queue:

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palmrun | Silent mode | Clear | Quit | Custom tag:

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PALM 23.04

# palmrungui

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Start Tools Help

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Input files for the selected job

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Browse at: /home/maronga/palm_23.04/JOBS Update (5 min)

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palmrun | Silent mode | Clear | Quit | Custom tag:

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User code for the selected job

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# palmrungui

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Monitoring files for the selected job

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Output files:

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# palmrungui

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## Output files for selected job

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2023/08/08 15:21: run_B2_sub035_upgrade100v2 (default)

Execution

Selected job: run_B2_sub035v2

Configuration: default

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Use pre-built source code

Cores: 4 | Tasks per node: 4

Wallclock time (s): | Queue:

Account: | Wait for job:

Mode: Initial run

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palmrun | Silent mode | Clear | Quit | Custom tag:

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PALM 23.04

# palmrungui

Use context dialog to create new / copy jobs

palmrungui

Start Tools Help

Default | Advanced

**History**

Recently submitted jobs:

2023/08/08 15:21: run_B2_sub035v2 (default)
2023/08/08 15:21: run_B2_sub035_upgrade100v2 (default)

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run_B2_sub035_upgrade100v2
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Output files:

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.directory
land_surface_model_mod.f90
radiation_model_mod.f90
surface_mod.f90
urban_surface_mod.f90

Monitoring files:

**Execution**

**Selected job:** run_B2_sub035v2

Configuration: default

Edit | (re-)Build | Full rebuild

Use pre-built source code

Cores: 4 | Tasks per node: 4

Wallclock time (s): | Queue:

Account: | Wait for job:

Mode: Initial run

Allow restarts | Use cyclic fill method

Keep temporary data | Read sky view factors from file

Write sky view factors to file

palmrun | Silent mode | Clear | Quit | Custom tag:

palmrun -r run_B2_sub035v2 -a "d3#" -X "4" -c "default" -T "4"

PALM 23.04

# palmrungui

Open selected files via context dialog ("right mouse click")

palmrungui

Start Tools Help

Default Advanced

History

Recently submitted jobs:

2023/08/08 15:21: run_B2_sub035v2 (default)
2023/08/08 15:21: run_B2_sub035_upgrade100v2 (default)

Execution

Selected job: run_B2_sub035v2

Configuration: default

Edit (re-)Build Full rebuild

Use pre-built source code

Cores: 4 Tasks per node: 4

Wallclock time (s): Queue:

Account: Wait for job:

Mode: Initial run

Allow restarts Use cyclic fill method

Keep temporary data Read sky view factors from file

Write sky view factors to file

Browser

Browse at: /home/marenga/palm_23.04/JOBS/ Update (5 min)

Available jobs:

run_B2_sub035_upgrade100v2
run_B2_sub035v2

Input files:

.directory (58 B)
run_B2_sub035v2_p3d (6 KB)
run_B2_sub035v2_p3dr (4 KB)

User code:

.directory
land_surface_model_mod.f90
radiation_model_mod.f90
surface_mod.f90
urban_surface_mod.f90

Monitoring files:

Output files:

palmrun Silent mode Clear Quit Custom tag:

palmrun -r run_B2_sub035v2 -a "d3#" -X "4" -c "default" -T "4"

PALM 23.04

# palmrungui

Settings for job execution

palmrungui

Start Tools Help

Default | Advanced

History

Recently submitted jobs:

2023/08/08 15:21: run_B2_sub035v2 (default)
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Browser

Browse at: /home/maronga/palm_23.04/JOBS/ Update (5 min)

Available jobs:

run_B2_sub035_upgrade100v2
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Input files:

.directory (58 B)
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Output files:

User code:

.directory
land_surface_model_mod.f90
radiation_model_mod.f90
surface_mod.f90
urban_surface_mod.f90

Monitoring files:

Execution

Selected job: run_B2_sub035v2

Configuration: default

Edit | (re-)Build | Full rebuild

Use pre-built source code

Cores: 4 Tasks per node: 4

Wallclock time (s): Queue:

Account: Wait for job:

Mode: Initial run

Allow restarts
Keep temporary data
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Read sky view factors from file
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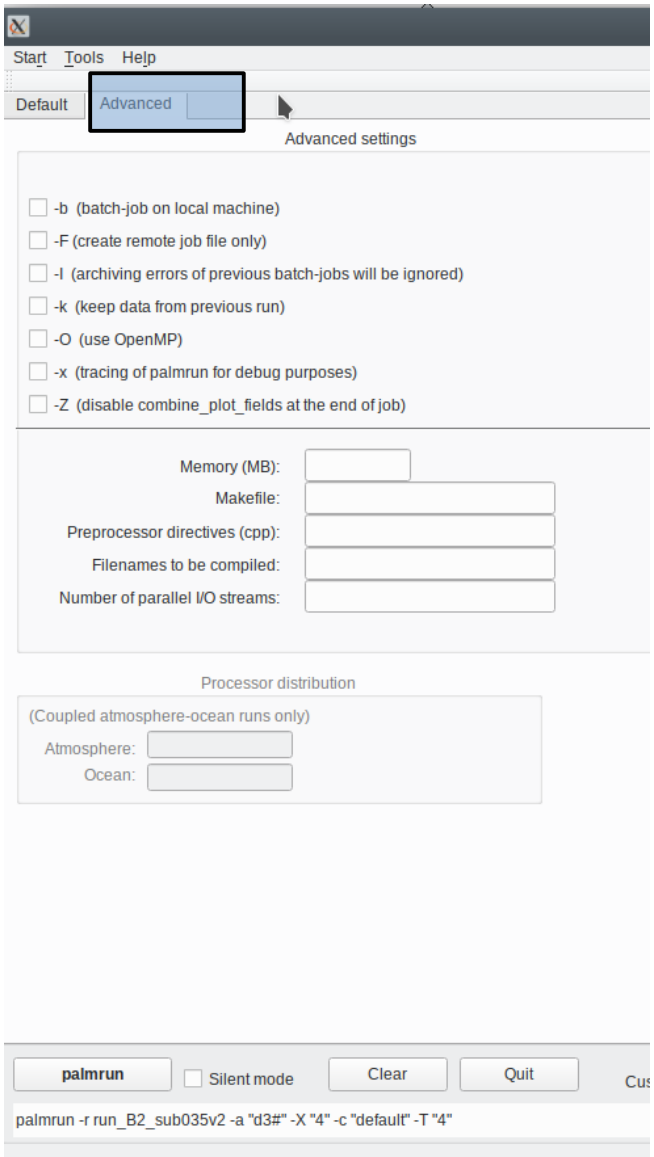
palmrun | Silent mode | Clear | Quit

Custom tag:

palmrun -r run_B2_sub035v2 -a "d3#" -X "4" -c "default" -T "4"

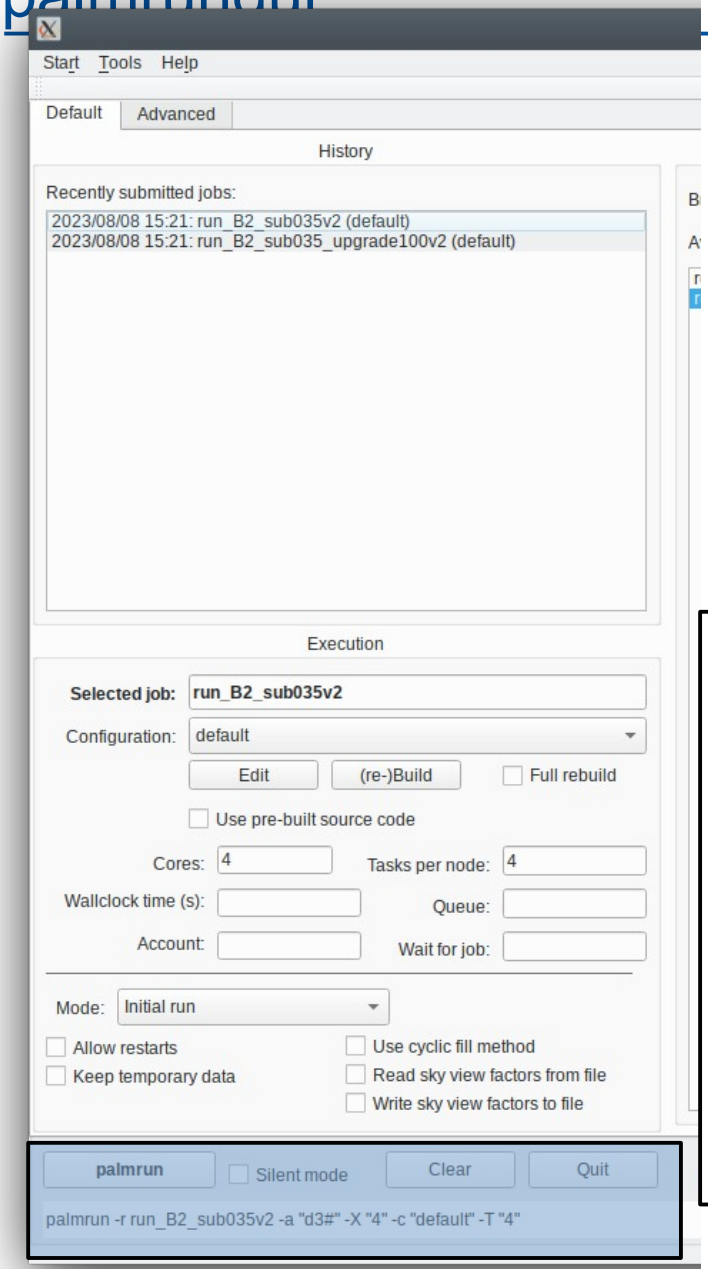
PALM 23.04

# palmrungui




- Advanced setting:
  be sure what you are doing…
  see `palmrun ?`

# palmrungui




- Execute PALM (= interactively or send job to queuing system) via button **palmrun**
- The bottom line shows the palmrun command. You could also directly execute this command in the shell!
- palmrun will be executed in seperate window (xterm):

```
 …
 *** creating executable and other sources for
the remote host
  *** nothing to compile for this run
  *** executable and other sources created

 *** submit the job


 --> palmrun finished

--> Press enter to continue...
```

# palmrungui

palmrungui

Start Tools Help

Default Advanced

Wait for results in output file list...

History

Recently submitted jobs:

2023/08/08 15:33: example_cbl (default)
2023/08/08 15:21: run_B2_sub035v2 (default)
2023/08/08 15:21: run_B2_sub035_upgrade100v2 (default)

Execution

Selected job: example_cbl

Configuration: default

Edit (re-)Build Full rebuild

Use pre-built source code

Cores: 4 Tasks per node: 4

Wallclock time (s): Queue:

Account: Wait for job:

Mode: Initial run

Allow restarts
Keep temporary data
Use cyclic fill method
Read sky view factors from file
Write sky view factors to file

Browser

Browse at: /home/maronga/palm_23.04/JOBS/ Update (5 min)

Available jobs:

example_cbl
run_B2_sub035_upgrade100v2
run_B2_sub035v2

Input files:

example_cbl_p3d (4 KB)

User code:

Monitoring files:

example_cbl_cpu.000 (3 KB)
example_cbl_header.000 (7 KB)
example_cbl_parin.000 (4 KB)
example_cbl_rc.000 (31 KB)
example_cbl_rc.001 (31 KB)
example_cbl_stdout.000 (10 KB)

Output files:

example_cbl_av_xz.000.nc (9 KB)
example_cbl_pr.000.nc (7 KB)
example_cbl_ts.000.nc (16 KB)
example_cbl_xy.000.nc (53 KB)
example_cbl_xz.000.nc (29 KB)

palmrun Silent mode Clear Quit Custom tag:

palmrun -r example_cbl -a "d3#" -X "4" -c "default" -T "4"

PALM 23.04

# palmrungui

Open files, e.g. via ncview

palmrungui

Start Tools Help

Default Advanced

History

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2023/08/08 15:21: run_B2_sub035v2 (default)
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Execution

Selected job: example_cbl

Configuration: default

Edit (re-)Build Full rebuild

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palmrun Silent mode Clear Quit Custom tag:

palmrun -r example_cbl -a "d3#" -X "4" -c "default" -T "4"

PALM 23.04

# palmrungui

Check monitoring files

palmrungui

Start Tools Help

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Edit (re-)Build Full rebuild

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palmrun Silent mode Clear Quit Custom tag:

palmrun -r example_cbl -a "d3#" -X "4" -c "default" -T "4"

PALM 23.04

## Advantages

- Extremely easy job mangement
- History of all submitted jobs (inculding failed ones)
- You can save a default configuration
- Quick overview of all related files for a job
- Works well for local and remote job submission
- Recommended during testing phase

## Limitations

- Output files will only be displayed if you run PALM locally or if you transfer data (automatically) to your local JOBS/OUTPUT directory (check .palm.iofiles configuration)
- Remote job chains do not work properly at the moment
  → Call of palmrungui might be required on compute host

# Advantages

**IMPORTANT NOTE:**
palmrungui has a bug in 25.04. and does not work for precursor runs and cyclic fill method.

If you want to work with palmrungui in Exercise 3, go to

`palm/[...]/packages/gui/palmrungui/`

and replace `./main.py` and `./palmrungui_files/mainwindow.ui`

With the two files available on seafile in the E-3 zip file!

(check .palm.iofiles configuration)

- Remote job chains do not work properly at the moment
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